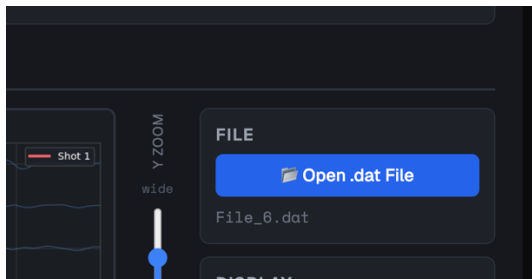


Seismic First Break Viewer

The viewer is a python widget that runs in a web page, and enables us to view the Tromino “.dat” files we saved in the field.

STEP 1

First we load our file:



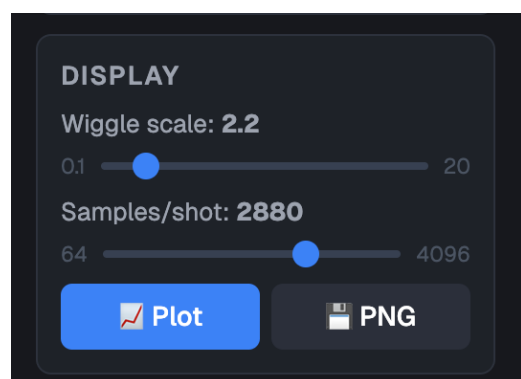
Select the correct file from your notes. You should see something like this:



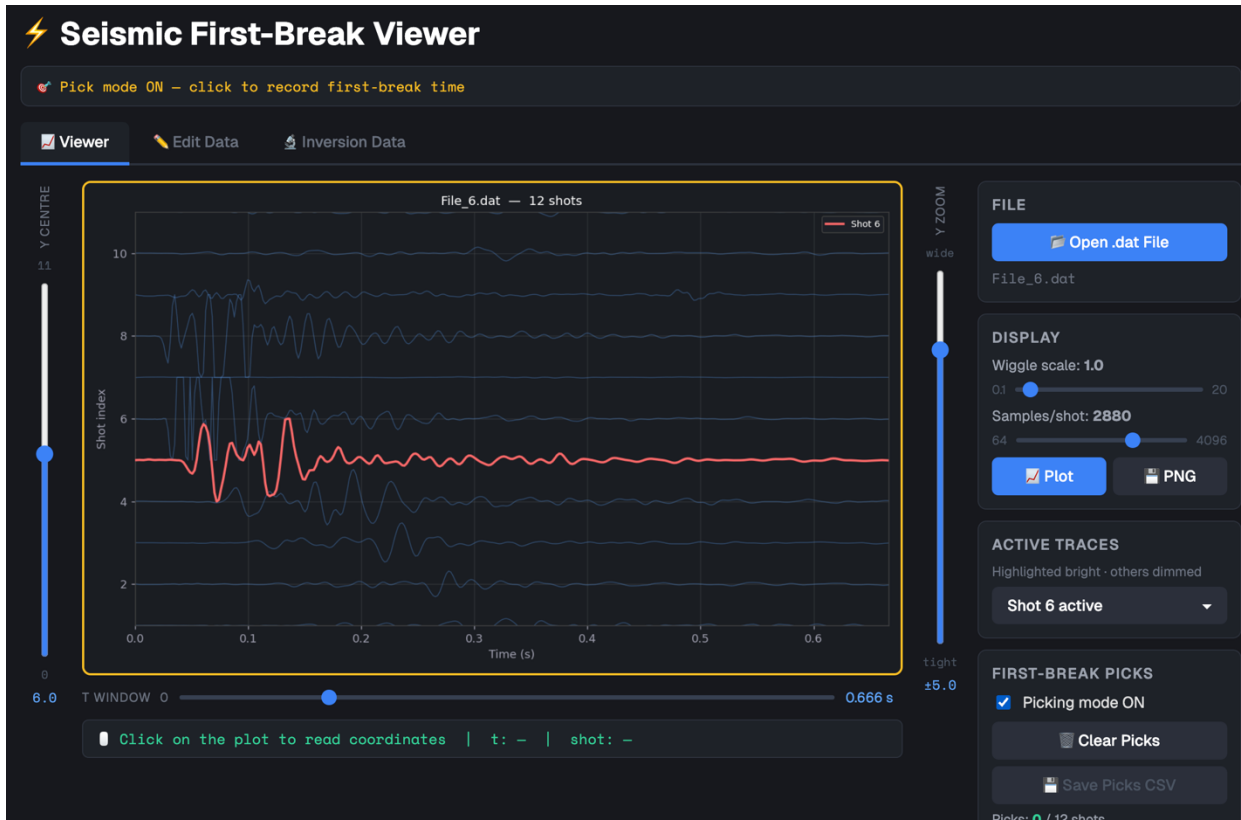
The number is on the y axis. The axis is the time since the trigger was activated. This should correspond to the shot. Be sure that you have identified all properly triggered shots. You should also know the receiver position, and the shot position, from your notes.

There are slides on the left, right, and bottom of the wiggle plot. These control your “Y centre” (ie. which shot is in the centre – enabling you to see all shots), the “Y zoom” zooming in or out of all the shots, and the Time window – you might want to shrink this to smaller amounts than the default (2s) to see the first arrivals better.

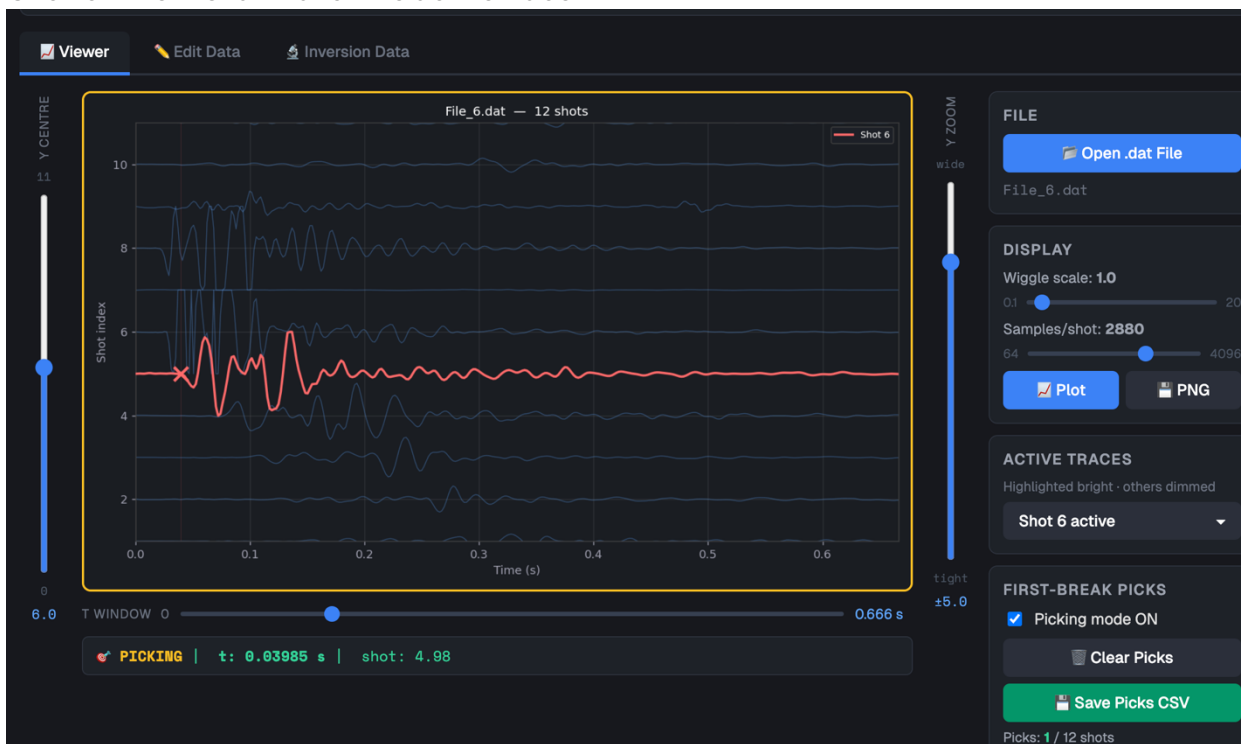
You can adjust the wiggle size with the slide below. This can make the shots near the receiver smaller, and the shots far away from the receiver bigger, to aid visibility.



Once you are happy with your view (you can change it as you go), you can then select a trace to make active (red one below), and then select the “Picking mode ON” box. Now we can pick first breaks. Note shot 6 is shot index # 5 here as python indexes start at “0”.

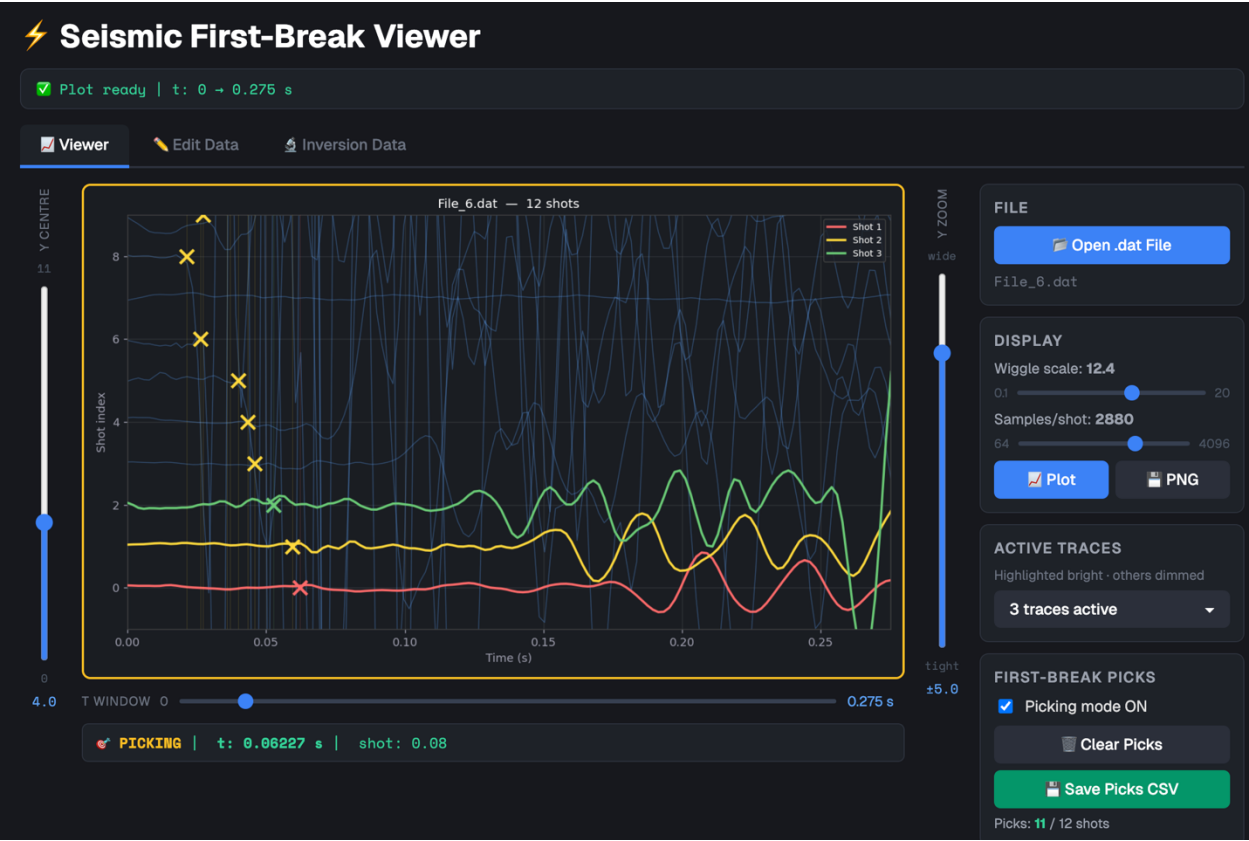


Click on the first arrival of the active trace:



You'll see a cross on the trace. Continue for all traces which have proper data. Note that time should always increase as you pick away from the receiver location.

Keep picking and adjusting the wiggles and Y centre. Note the first arrival will be MUCH smaller than the surface waves. Be careful when picking, and also watch out for noise.



Your data will be shown at the bottom. If you've made a mistake, you can delete that pick here:

Trigger count (0-indexed)
3

FILE SUMMARY
File_6.dat Samples: 165,888 Duration: 323.998 s Shots: 12

FIRST-BREAK PICKS TABLE

Shot #	Sample idx	Pick time (s)	Action
Shot 1	21171	0.062268	✕
Shot 2	42820	0.059484	✕
Shot 3	61335	0.052723	✕
Shot 4	73092	0.045767	✕
Shot 5	87104	0.043358	✕
Shot 6	99524	0.039848	✕
Shot 7	111256	0.026233	✕
Shot 9	132589	0.021371	✕
Shot 10	143674	0.027206	✕
Shot 11	152521	0.035958	✕
Shot 12	160920	0.036930	✕

When you are happy, click the “Save picks as CSV” button, and watch out for the save box (it might pop up behind the window).

STEP 2

Now go to the “Edit Data” tab. Here’s where you’ll enter the shot locations and receiver spot.

Seismic First-Break Viewer

Picks saved → /Users/oneillc5/Desktop/QUT_teaching/2026/ERB302/Fieldwork/firstbreaks.csv

Viewer

Edit Data

Inversion Data

Select “Open CSV” and load your data (this can be a bit touchy, just keep trying). You’ll then see this:

⚡ Seismic First-Break Viewer

✓ Loaded for editing: firstbreaks.csv

Viewer

Edit Data

Inversion Data

LOAD PICKS CSV

Open CSV

firstbreaks.csv

GEOMETRY — APPLY TO SELECTED

Select rows then fill and apply, or edit cells directly.

Shot X (m)

e.g. 44.0

Shot Y (m / elev)

0

Receiver X (m)

e.g. 50.0

Receiver Y (m / elev)

0

✓ Apply to Selected

Apply to All

☐	Shot #	Sample	Time (s)	Shot X	Shot Y	Recv X	Recv Y
☐	1	21171	0.06226824973164539	—	—	—	—
☐	2	42820	0.05948416714731932	—	—	—	—
☐	3	61335	0.052722823728241726	—	—	—	—
☐	4	73092	0.04576687300953655	—	—	—	—
☐	5	87104	0.04335757008252552	—	—	—	—
☐	6	99524	0.03984783929738562	—	—	—	—
☐	7	111256	0.026233087484385156	—	—	—	—
☐	9	132589	0.02137067612259929	—	—	—	—
☐	10	143674	0.02720556975674234	—	—	—	—
☐	11	152521	0.03595791020795692	—	—	—	—
☐	12	160920	0.03693039248031409	—	—	—	—

Since the receiver is at the same place each time, we can select all the shots (box at top next to “Shot #”). Type in the receiver location in the left, then apply to selected:

⚡ Seismic First-Break Viewer

✓ Applied geometry to 11 row(s)

Viewer Edit Data Inversion Data

LOAD PICKS CSV

Open CSV

firstbreaks.csv

GEOMETRY — APPLY TO SELECTED

Select rows then fill and apply, or edit cells directly.

Shot X (m)

e.g. 44.0

Shot Y (m / elev)

0

Receiver X (m)

45

Receiver Y (m / elev)

0

✓ Apply to Selected

Apply to All

☒	Shot #	Sample	Time (s)	Shot X	Shot Y	Recv X	Recv Y
☒	1	21171	0.06226824973164539	—	0	45	0
☒	2	42820	0.05948416714731932	—	0	45	0
☒	3	61335	0.052722823728241726	—	0	45	0
☒	4	73092	0.04576687300953655	—	0	45	0
☒	5	87104	0.04335757068252552	—	0	45	0
☒	6	99524	0.03984783929738562	—	0	45	0
☒	7	111256	0.026233087484385156	—	0	45	0
☒	9	132589	0.02137067612259929	—	0	45	0
☒	10	143674	0.02720556975674234	—	0	45	0
☒	11	152521	0.03595791020795692	—	0	45	0
☒	12	160920	0.03693039248031409	—	0	45	0

You can manually enter the shot x location for each shot you have digitized.

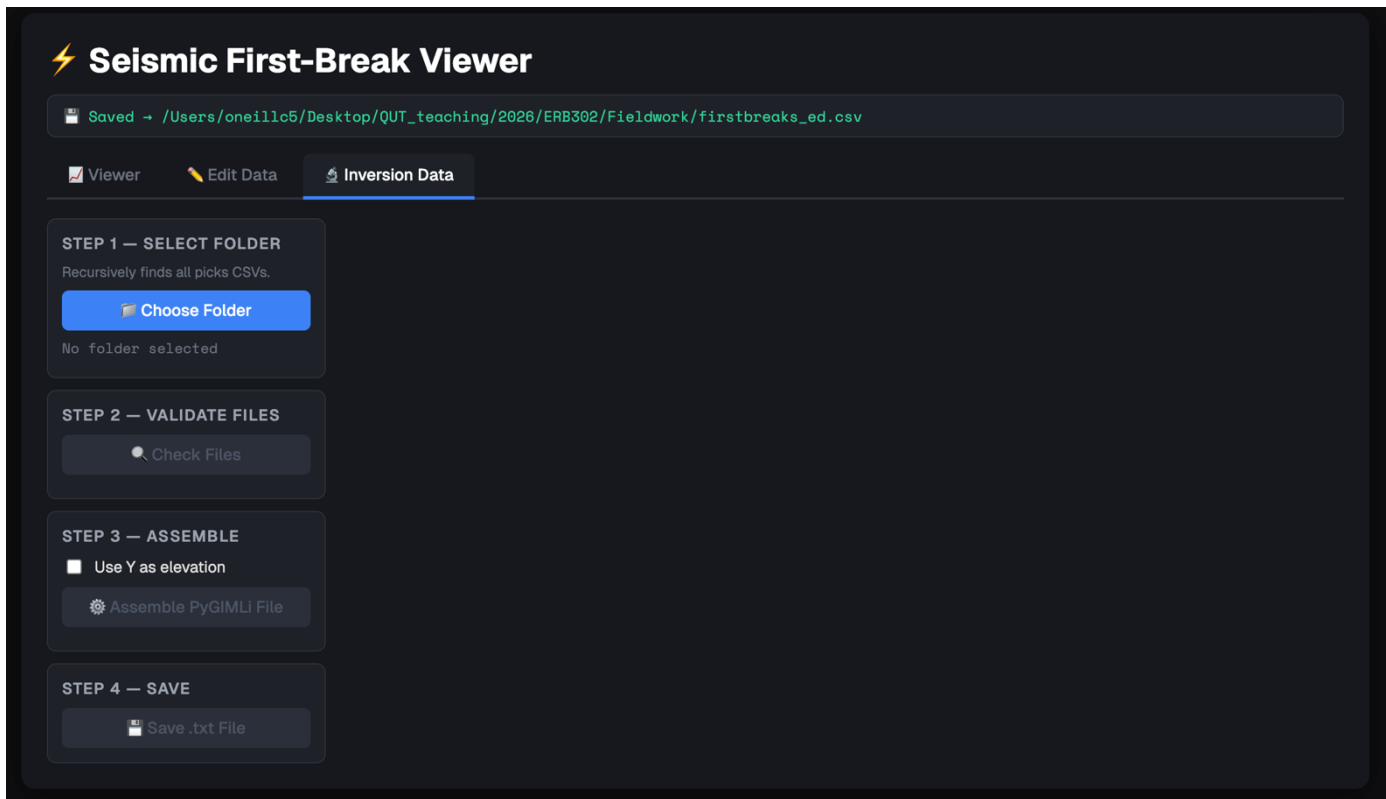
Lastly, you can extract the shot height from QGIS along the line. You need to use the 1m LIDAR from the survey from ELVIS. You can construct a line with the end points from the DGPS data provided (load as delimited text). Then you can sample every 5m along the line for heights. Then enter these here.

Save your edited csv. Note you should keep the filename number of the original “dat” file as you go, so you can keep track.

STEP 3

Once you have finished with your individual csvs and they have all their positions, you can develop an inversion file. Click on the “Inversion Data” tab.

This assumes you have all you “csv” files in the one folder. Make sure they are and then select it.



Once the folder is loaded, try “Step 2” to validate the csv files.

The third step is to convert all these different files into a format that pygimli will understand. If you have included elevation data from the lidar DEMS into your individual files, then tick that box (else they will be 0m, and the inversion will ignore the elevation change. This can lead to some weird velocity anomalies).

Fourth, save the final ready for inversion in pygimli.